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**DEPARTMENT OF INFORMATION
TECHNOLOGY**

OPERATING SYSTEM MANAGEMENT

Process • Memory • Storage • Security • Device Management

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Course/Section:

BSCS-3A

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Module Overview

An operating system does more than provide a desktop or application interface. It acts as the central manager of computer resources. Every time a user opens an application, saves a file, connects a device, logs in, or runs several programs at once, the operating system coordinates the underlying resources required to make those actions work.

This module develops a practical understanding of five major operating system management functions: process management, memory management, storage management, security management, and device management. The discussion then compares how these functions appear in Windows and Linux.

Learning Outcomes

At the end of the module, students should be able to:

1. Explain the role of process management and describe scheduling, synchronization, and inter-process communication.
2. Describe how an operating system allocates RAM and uses virtual memory, paging, and segmentation.
3. Explain file system and mass-storage management, including common file systems and disk scheduling.
4. Relate confidentiality, integrity, and availability to operating system security mechanisms and best practices.
5. Explain how device drivers, I/O management, Plug and Play, and device scheduling allow software to interact with hardware.
6. Compare selected operating system management features in Windows and Linux.
7. Apply operating-system concepts to practical scenarios through observation, analysis, and problem-solving activities.

Key Terms

Term	Working Definition
Process	A program that is currently executing.
Scheduling	The OS decision process for assigning CPU time to processes.
Synchronization	Coordination that prevents processes from interfering when they share resources.
IPC	Mechanisms that allow processes to communicate safely.

Virtual Memory	A method that uses secondary storage to extend the apparent amount of usable memory.
Paging	Memory organization using fixed-size blocks called pages.
Segmentation	Memory organization using variable-sized logical segments.
File System	A method used by the OS to organize and access stored data.
Authentication	Verification of a user's identity before granting access.
Access Control	Rules that determine who may use specific system resources.
Encryption	Transformation of data into an unreadable form without the correct key.
Device Driver	Software that translates OS requests into hardware-specific operations.
Plug and Play	Automatic device detection and configuration by the operating system.

Diagnostic Questions

Before reading the lesson, answer briefly. Your answers are not graded; they are intended to activate prior knowledge.

8. What happens inside a computer when two or more applications are open at the same time?

Answer: The OS goes through a process called multitasking

9. Why can a computer become slow when available RAM is low?

Answer: A computer can become slow for several reasons; one of these reasons is when there is limited workspace. RAM acts as temporary storage; when it becomes full, it can no longer store new data, and it will rely on the SSD.

10. Why do USB drives, hard drives, and SSDs need a file system?

Answer: USB drives, hard drives, and SSDs need a file system to organize data into files and folders for easy access.

11. What operating system features help protect a computer from unauthorized access?

Answer: Operating systems protect computers from unauthorized access through features such as user authentication, file permissions, encryption, firewalls, and many more.

12. Why does some hardware fail to work properly when a driver is missing or incompatible?

Answer: Hardware fails to work properly without the correct driver because drivers facilitate communication between the operating system and hardware, so when the driver is missing or incompatible, the OS cannot communicate with the hardware.

Activity 1 — Process Observation and Classification

Purpose: Observe real processes and connect them with process-management concepts.

Materials/Tools: Windows Task Manager or an equivalent process-monitoring tool; worksheet.

Procedure:

13. Open at least three applications, such as a browser, text editor, and media application.
14. Open the system process-monitoring tool and identify the processes related to the applications.
15. Record the process/application name, visible CPU activity, and visible memory use.
16. Classify each process as primarily user-facing or background/system related.
17. Choose one process and explain what might happen if the OS stopped scheduling it.
18. Write a short reflection explaining why process isolation is useful when one application crashes.

Required Output: A completed observation table plus a 150–200 word reflection.

A. Process Observation Table

System Information Observed:

- **Total CPU usage:** 34%
- **Total memory usage:** 95%
- **Total disk usage:** 6%
- **Total network usage:** 0%

Application/Process	Process Name	Visible CPU Activity	Visible Memory Use	Classification
Microsoft Edge	Microsoft Edge (14)	8.5%	864.4 MB	User-facing
Microsoft Word	Microsoft Word	1.1%	84.2 MB	User-facing
Roblox	Roblox (2)	10.2%	301.3 MB	User-facing
Task Manager	Task Manager	1.3%	64.2 MB	Background/system-related

Observation:

Among the listed applications, Microsoft Edge used the most memory at 864.4 MB, followed by Roblox at 301.3 MB, while Microsoft Word used the least. In terms of CPU usage, Roblox took up 10.2% of the CPU, Microsoft Edge used 8.5%, and Microsoft Word only used 1.1%. During my observation, I noticed that CPU usage changed a lot, going up and down based on what the

applications were doing. I also saw that memory usage gradually increased the longer the applications were open. When I took the screenshot, the computer was at 34% CPU usage and 95% memory usage.

Figure 1. Windows Task Manager Processes Observation

Processes

Run new task

End task

Eff

Name	Status	34% CPU	95% Memory	6% Disk	0% Network
Apps (4)					
>  Microsoft Edge (14)		8.5%	846.4 MB	7.7 MB/s	0.1 Mbps
>  Microsoft Word		1.1%	84.2 MB	0 MB/s	0 Mbps
>  Roblox (2)		10.2%	301.3 MB	0 MB/s	0 Mbps
>  Task Manager		1.3%	64.2 MB	0.1 MB/s	0 Mbps

B. Process Scheduling Analysis

Chosen process: Roblox

If the operating system stopped scheduling the Roblox process, it would no longer receive CPU time to perform its tasks. As a result, the game might freeze, stop responding, or fail to continue running properly. Other applications, such as Microsoft Word and Microsoft Edge, could continue receiving CPU time.

During my observation, I noticed that CPU usage constantly changed from high to low and then increased again. This shows that processes do not always require the same amount of CPU time. Process scheduling allows the operating system to manage these changing demands and distribute CPU time among running applications.

C. Reflection (150–200 words)

During my observation, I saw that Microsoft Edge, Microsoft Word, Roblox, and Task Manager were running at the same time. The Task Manager showed 34% CPU usage and 95% memory usage. Microsoft Edge used the most memory among the listed applications, with 846.4 MB.

I also noticed that CPU usage constantly changed from high to low and then increased again. This may happen because applications perform different tasks at different times. I observed that memory usage also increased the longer my applications remained open.

Process scheduling is important because the operating system must manage the CPU time given to different processes. This allows several applications to run at the same time. Memory management is also important because the operating system allocates RAM to running processes and manages available memory.

Process isolation helps maintain system stability. If one application crashes, the operating system can manage the affected process separately, allowing other applications to continue running. Overall, my observation helped me understand how process and memory management affect the performance of a computer.

References

Cavite State University, Department of Information Technology. (n.d.). *Module 3. Operating System Management* [Course module].

Activity 2 — Memory Pressure Investigation

Purpose: Relate RAM use and virtual memory to system performance through controlled observation.

Materials/Tools: Computer with system performance monitor; 3–4 ordinary applications.

Procedure:

19. Record the current memory usage before opening additional applications.
20. Open several applications one at a time. After each application, observe the change in memory use.
21. Identify which application uses the most memory.
22. Observe whether the system becomes less responsive as memory use increases.
23. Explain the role of virtual memory if physical RAM becomes insufficient.
24. Answer: How does memory management affect a system with limited RAM? Give evidence from your observation.

Required Output: A memory-observation table and a short analysis of 200–250 words.

A. Memory-Observation Table

System Information Observed:

- **Total installed RAM:** 8.0 GB DDR4
- **Memory in use:** 7.4 GB
- **Available memory:** 289 MB
- **Memory usage shown in Task Manager:** 94%

Memory Usage of Running Applications		
Application	CPU Usage	Memory Usage
Microsoft Edge (15)	0.6%	656.2 MB
Microsoft Word	0%	76.1 MB
RobloxPlayerBeta	3.5%	412.6 MB
Task Manager	1.7%	84.5 MB
Windows Explorer	0.4%	137.6 MB

Observation: Among the applications shown, Microsoft Edge used the most memory at 656.2 MB.

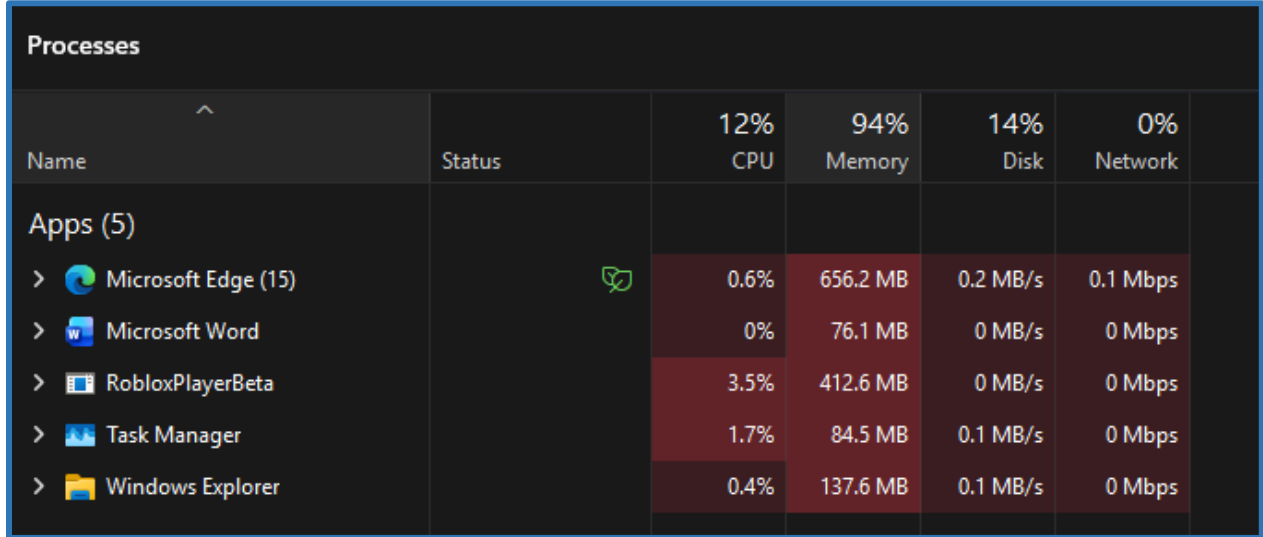
Memory-Observation Record				
Observation Stage	Applications Open	Total Memory Usage	Available Memory	System Response
Previous reading	Microsoft Edge, Microsoft Word, and Roblox	95%	Not recorded	The system response and available memory were not recorded during the previous observation because I checked only total memory usage and application memory values. Still, the computer remained responsive while

				the applications were open.
New reading	Microsoft Edge, Microsoft Word, Roblox, and Windows Explorer	94%	289 MB	The computer remained responsive while the applications were open, although memory usage was high.

Figure 1. Windows Task Manager Memory Observation



Figure 2. Windows Task Manager Processes Observation



The screenshot shows the Windows Task Manager 'Processes' tab. At the top, summary statistics are displayed: 12% CPU, 94% Memory, 14% Disk, and 0% Network. Below this, a list of applications is shown under the heading 'Apps (5)'. Each application row includes an expandable icon, the application name, a status icon, and columns for CPU, Memory, Disk, and Network usage.

Name	Status	12% CPU	94% Memory	14% Disk	0% Network
Apps (5)					
> Microsoft Edge (15)		0.6%	656.2 MB	0.2 MB/s	0.1 Mbps
> Microsoft Word		0%	76.1 MB	0 MB/s	0 Mbps
> RobloxPlayerBeta		3.5%	412.6 MB	0 MB/s	0 Mbps
> Task Manager		1.7%	84.5 MB	0.1 MB/s	0 Mbps
> Windows Explorer		0.4%	137.6 MB	0.1 MB/s	0 Mbps

Note: Task Manager was open in the second screenshot for monitoring purposes.

B. Memory Usage Comparison

Application	Previous Reading	New Reading
Microsoft Edge	846.4 MB	656.2 MB
Microsoft Word	84.2 MB	76.1 MB
Roblox	301.3 MB	412.6 MB
Task Manager	64.2 MB	84.5 MB
Windows Explorer	Not open	137.6 MB

Observation: During my observation, I noticed that CPU and memory usage changed while my applications were running. CPU usage increased and decreased depending on the activities of the applications. I also observed that memory usage could increase the longer applications remained open. In my previous reading, total memory usage was 95%, while my new reading showed 94%. Although the total memory usage decreased slightly, Roblox's memory usage increased from 301.3 MB to 412.6 MB.

C. Short Analysis (200–250 words)

Memory management is an important function of the operating system because it controls how RAM is allocated and used by running applications. It allows several programs to operate at the same time while managing the available memory.

During my observation, I recorded the memory usage of my computer using Windows Task Manager. In my previous reading, the total memory usage was 95%, while my new reading showed 94%. The computer had 8.0 GB of RAM, with 7.4 GB in use and 289 MB available in the new reading. These results show that most of the computer's memory was being used.

Among the applications in my new observation, Microsoft Edge used the most memory at 656.2 MB, followed by Roblox at 412.6 MB. I also noticed that CPU and memory usage changed while my applications were running. The CPU usage increased and decreased depending on the activities of the applications. I observed that memory usage could also increase the longer applications remained open.

This investigation shows that effective memory management supports multitasking. It allows the computer to continue operating even when RAM is limited, but performance may decrease as more data must be transferred between RAM and storage. The operating system must therefore allocate memory carefully and manage active and inactive data efficiently.

References

Cavite State University, Department of Information Technology. (n.d.). *Module 3. Operating System Management* [Course module].

Activity 3 — Storage Decision Case Study

Purpose: Compare storage choices and apply file-system and storage-management concepts.

Materials/Tools: Worksheet; access to basic storage information on a laboratory computer.

Procedure:

25. Inspect the file system used by the computer's main drive and one removable drive, if available.
26. Record the storage type, capacity, and file system that can be observed.

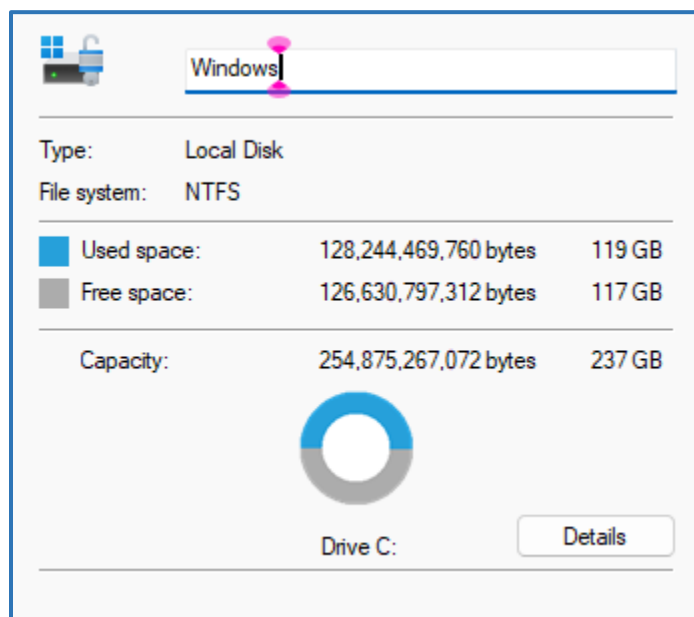
27. Create a scenario in which a school must choose between an HDD and SSD for a computer laboratory.
28. List at least three decision factors, including speed, capacity, cost, and expected use.
29. Explain how file systems and disk scheduling contribute to data access.
30. Conclude which storage option you would choose for the scenario and justify your decision.

Required Output: A one-page storage recommendation with evidence-based justification.

A. Storage Observation

Application	Storage Type	Capacity	File System
Main Drive (C:)	SSD	237 GB	NTFS

Figure 1. Main Storage Drive Information



B. Storage Decision Scenario

A school plans to upgrade the computers in its laboratory. The computers are used for programming, browsing, document editing, multimedia activities, and storing student files. The school must choose between a **Hard Disk Drive (HDD)** and a **Solid-State Drive (SSD)**.

C. Decision Factors

Factor	HDD	SSD
Speed	Slower because it uses moving parts	Faster because it uses flash memory
Capacity	Usually offers larger capacity at a lower price	Usually more expensive for the same capacity
Cost	More affordable	More expensive
Durability	More sensitive to physical movement	More resistant to physical shock
Expected Use	Suitable for large file storage and backups	Suitable for fast booting, applications, and multitasking

D. Storage Recommendation

I recommend using an **SSD** for the school computer laboratory. SSDs provide faster booting, quicker application loading, and better overall system responsiveness. This is useful because students will regularly open programming software, web browsers, documents, and other applications.

Although SSDs may cost more than HDDs, their speed and durability make them a practical choice for computers used every day. The school may also use an HDD or network storage for backups and storing large student files.

File systems such as **NTFS** help organize and manage files on a storage device. They also support features such as file permissions and large file handling. Disk scheduling helps organize requests for reading and writing data so that the operating system can access storage efficiently.

Overall, an SSD is the better choice for the laboratory's main drive because it improves performance and reduces waiting time. An HDD may still be used as an additional storage or backup device.

References

Cavite State University, Department of Information Technology. (n.d.). *Module 3. Operating System Management* [Course module].

Activity 4 — Operating System Security Audit

Purpose: Identify basic security controls and recommend improvements without changing protected system settings.

Materials/Tools: Computer or provided screenshots; audit worksheet.

Procedure:

31. Check whether automatic system updates are enabled or identify where update status is shown.
32. Identify the authentication method used on the device (for example, password, PIN, or biometric).
33. Locate the operating system firewall status.
34. Identify whether drive/device encryption is available or enabled, if visible.
35. For each control, explain whether it primarily supports confidentiality, integrity, availability, or more than one objective.
36. Recommend three realistic security improvements for a student or school computer.

Required Output: A completed security-audit checklist plus three recommendations.

A. Security-Audit Checklist

Security Control	What to Check	Observed Status	CIA Objective Supported	Evidence
Automatic System Updates	Check whether Windows Update is enabled or whether updates are available.	Windows Update shows that a restart is required. A Microsoft Defender Antivirus intelligence update is installing, and the 2026-09 Security Update is pending restart.	Integrity and Availability	Figure 1

Authentication Method	Identify whether the device uses a password, PIN, fingerprint, or other method.	Windows Settings shows PIN (Windows Hello) as an available sign-in method. A security key option is also available.	Confidentiality and Integrity	Figure 2
Firewall Status	Check whether Microsoft Defender Firewall is turned on.	Microsoft Defender Firewall is on for the active public network.	Confidentiality, Integrity, and Availability	Figure 3
Device Encryption	Check whether BitLocker or device encryption is available or enabled	Device encryption is on . The screen also shows BitLocker drive-encryption settings.	Confidentiality and Integrity	Figure 4

Figure 1. Windows Update Status of the Computer

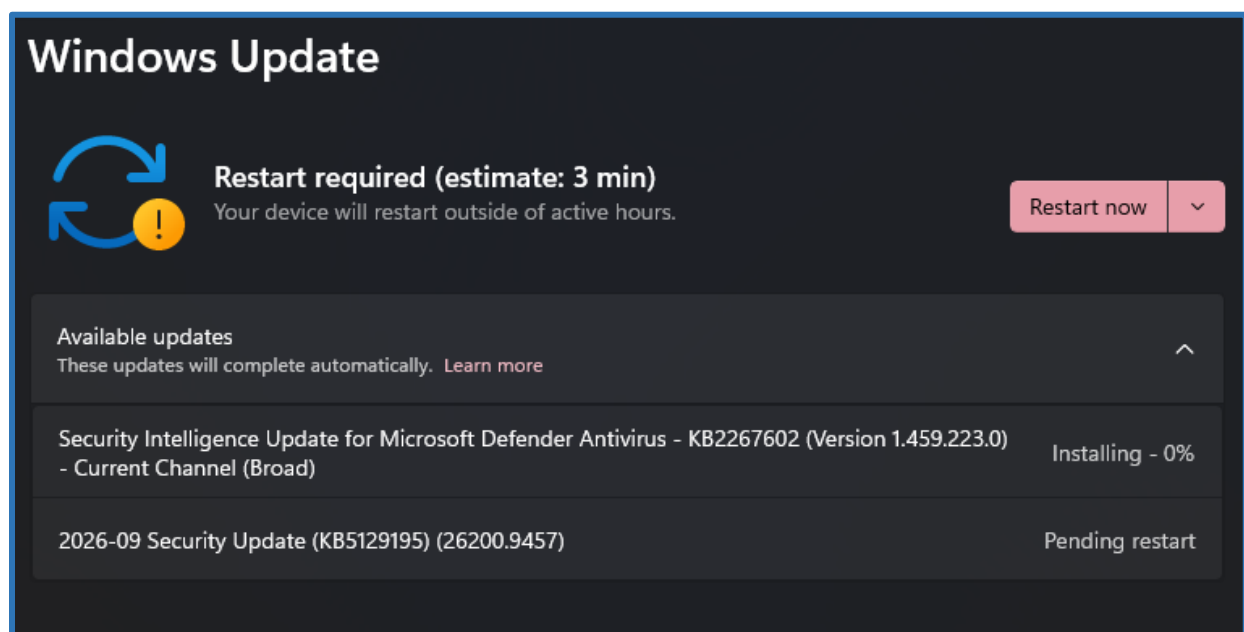


Figure 2. Authentication Methods Available On the Computer

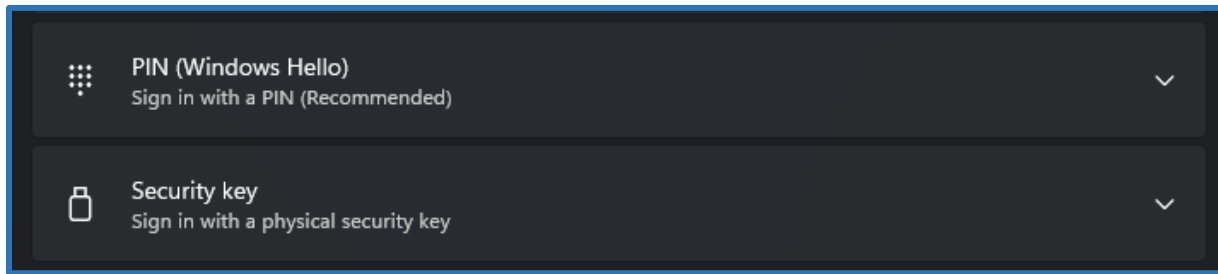


Figure 3. Windows Defender Firewall Status

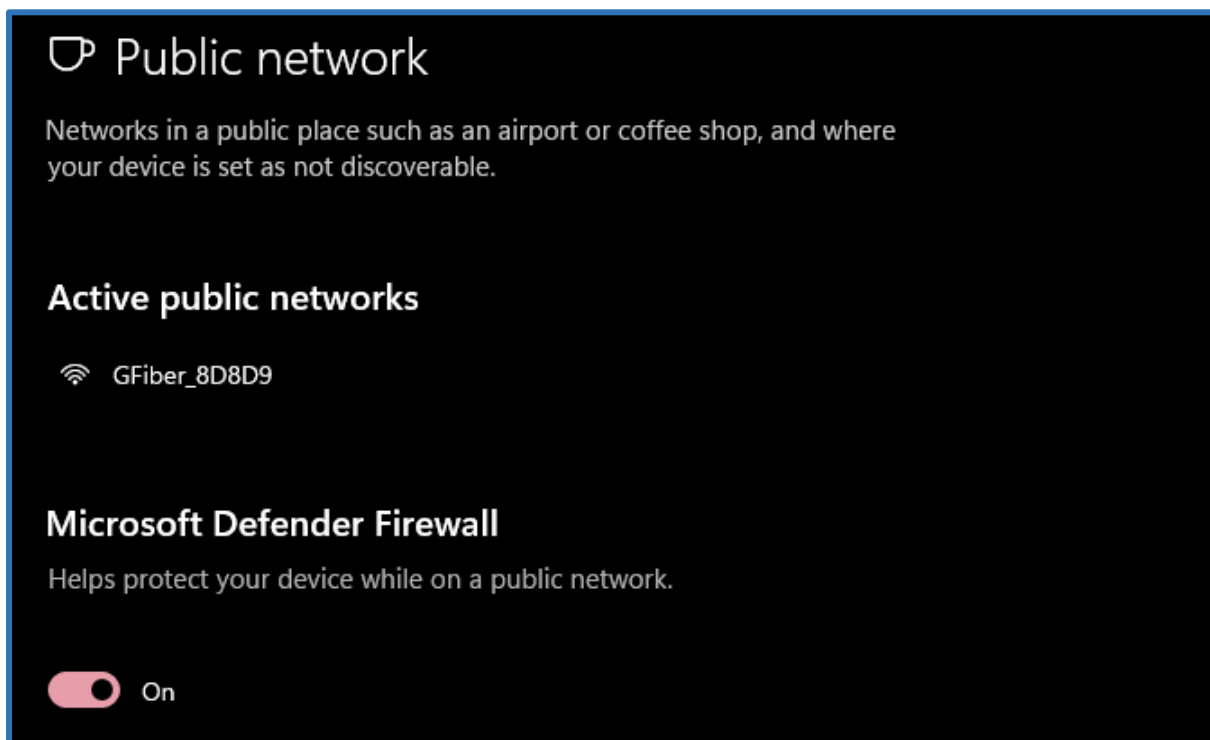


Figure 4. Device Encryption Status



B. Security-Audit Findings

The computer has several important security protections enabled. Device encryption is turned on, and Microsoft Defender Firewall is active on the public network. These controls help protect files and reduce the risk of unauthorized network access.

The computer also has Windows updates available. Since the screen shows **“Restart required”** and an update is **“Pending restart,”** the updates should be completed by restarting the computer when it is convenient and safe to do so.

The available authentication methods include a Windows Hello PIN and a security key. These methods help verify the user’s identity before access is granted.

C. Three Recommended Security Improvements

1. Complete pending Windows updates

The user should restart the computer to finish the pending security updates. Regular updates help repair known security weaknesses and keep the operating system protected.

2. Use a strong authentication method

The user should use a secure PIN or password and avoid sharing it with others. Multi-factor authentication should also be enabled when available.

3. Keep firewall and encryption enabled

Microsoft Defender Firewall and device encryption should remain enabled. The firewall helps control network connections, while encryption protects stored files from unauthorized access.

References

Cavite State University, Department of Information Technology. (n.d.). *Module 3. Operating System Management* [Course module].

Activity 5 — Hardware-to-OS Interaction Map

Purpose: Trace how a user action moves through applications, the operating system, drivers, and hardware.

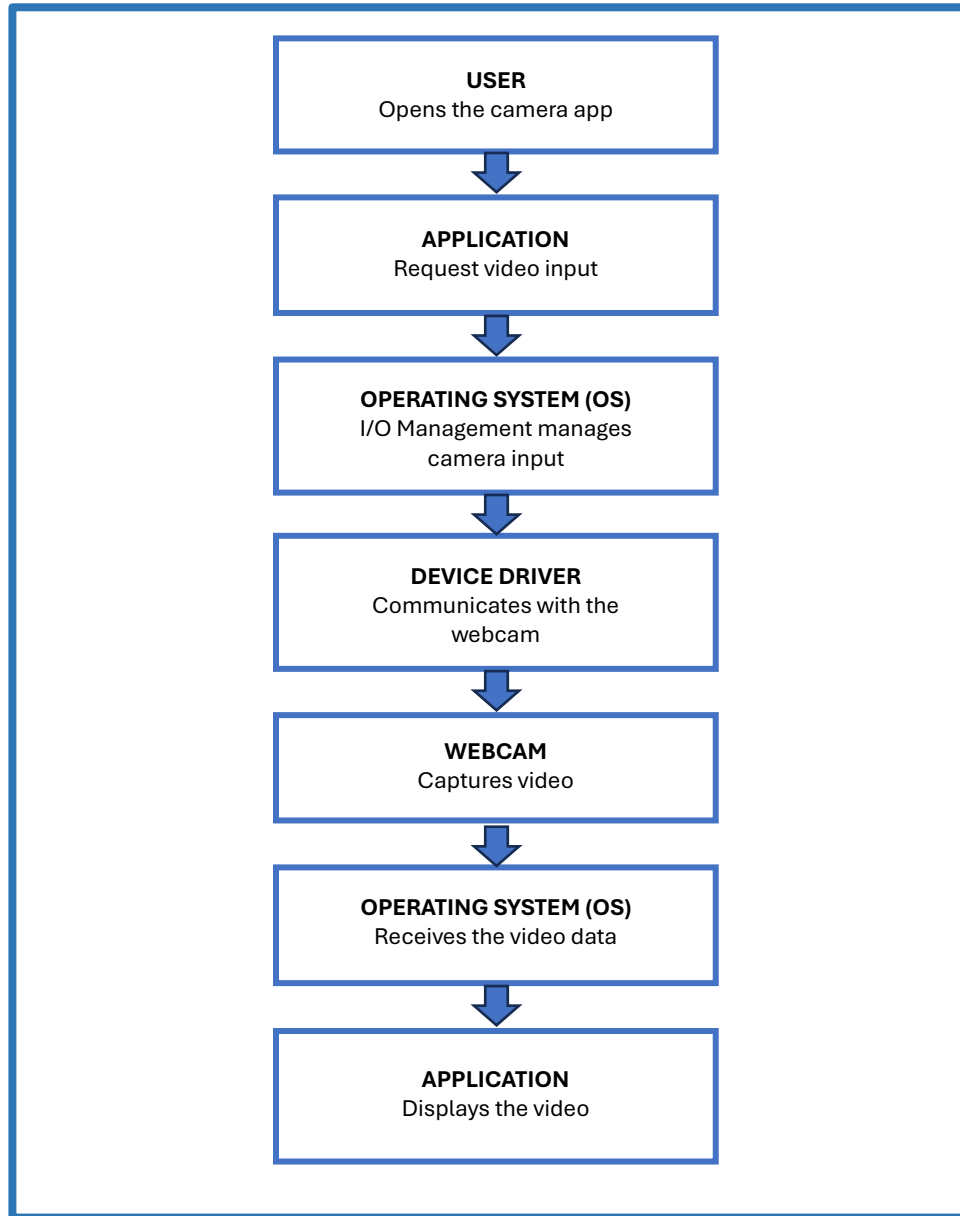
Materials/Tools: Paper or diagramming application.

Procedure:

37. Choose one device: printer, USB storage, keyboard, webcam, or external drive.
38. Draw a flow beginning with a user action and ending with the hardware response.
39. Include at least these components: user, application, operating system, device driver, and hardware device.
40. Add labels showing where input/output management occurs.
41. Add a note explaining how Plug and Play might be involved.
42. Present the diagram to a partner and revise any unclear step.

Required Output: One labeled interaction diagram with a 100–150 word explanation.

A. Interaction Diagram



B. Plug and Play

Plug and Play allows the operating system to detect and configure compatible hardware automatically. When a compatible webcam is connected, the operating system can recognize the device and load or install the appropriate driver when necessary.

C. Explanation (100–500 words)

When a user opens a camera application, the application requests video input from the webcam. The operating system manages this request through its Input/Output (I/O) management function. The device driver provides communication between the operating system and the webcam by translating the required commands into a form the hardware can understand. The webcam then captures video and sends the input back to the computer. The operating system manages the incoming data and allows the application to receive and display the video. This interaction allows the user to use the webcam without directly controlling its electronic components. Plug and Play can also help when a compatible webcam is connected because the operating system can detect the device and configure the necessary driver. This allows the webcam to communicate properly with the operating system and application.

References

Cavite State University, Department of Information Technology. (n.d.). *Module 3. Operating System Management* [Course module].

Activity 6 — Windows vs. Linux Comparison Matrix

Purpose: Synthesize the five management areas by comparing two operating system environments.

Materials/Tools: Module notes and comparison worksheet.

Procedure:

43. Create a five-row matrix for process, memory, storage, security, and device management.
44. For each row, summarize the Windows example in one or two sentences.
45. Summarize the Linux example in one or two sentences.
46. Identify one similarity in purpose between the two operating systems.
47. Identify one difference in the tool, mechanism, or terminology used.
48. Write a concluding paragraph explaining why different operating systems still need the same management functions.

Required Output: Completed comparison matrix and 150-word conclusion.

A. Comparison Matrix

Management Area	Windows	Linux	Similarity	Difference
Process Management	Windows manages running programs and allows users to monitor or stop them through tools such as Task Manager.	Linux manages running processes through its scheduling system and allows multiple processes to work at the same time.	Both control running programs and help distribute CPU time.	
Memory Management	Windows can use virtual memory to help applications continue running when the available RAM becomes limited.	Linux can use swap space as additional memory when the physical RAM is not enough.	Both help manage limited RAM and support multitasking.	
Storage Management	Windows organizes files on drives using file systems such as NTFS and can support FAT32 on some external devices.	Linux can use file systems such as EXT4 and provides permissions for controlling access to files.	Both organize stored files and control how data is accessed.	
Security Management	Windows provides features such as	Linux can use security tools such as SELinux and	Both use security mechanisms to	

	Windows Defender, BitLocker, and Windows Hello to help protect the system and its data.	firewall utilities to control and protect system resources.	prevent unauthorized access and protect data.	
Device Management	Windows can recognize and manage connected hardware using Device Manager and Plug and Play.	Linux can detect and configure devices using udev .	Both allow software and the operating system to communicate with hardware.	

B. Conclusion (150 words)

Windows and Linux may look and work differently, but they both have to perform important management tasks for a computer to function properly. Both operating systems manage programs, memory, storage, security, and connected devices. However, the way they perform these tasks is not exactly the same. Windows provides user-oriented tools such as Task Manager and Device Manager, while Linux uses mechanisms such as its process scheduler and udev. Their storage and memory systems also use different approaches, such as NTFS and virtual memory in Windows and EXT4 and swap space in Linux. Security features also differ, with Windows offering tools such as BitLocker and Windows Defender, while Linux can use SELinux and firewall utilities. Despite these differences, the purpose remains similar: the operating system manages computer resources, protects data, and helps applications communicate with hardware. This shows that different operating systems can use different methods while still performing the same basic responsibilities.

Guided Discussion

49. How does memory management affect performance on a computer with limited RAM? Use virtual memory in your explanation.

Answer: Memory management affects how well a computer can run several applications at the same time. When RAM becomes limited, the operating system can use **virtual memory** by temporarily moving some less-used data from RAM to secondary storage. This allows the computer to continue running, but it may become slower because storage is slower than RAM.

50. What are the trade-offs between SSD and HDD storage when choosing primary storage for a computer?

Answer: An **SSD** provides faster data access and can make a computer feel more responsive. An **HDD** can provide larger storage capacity at a lower cost. The choice depends on factors such as speed, capacity, cost, and how the computer will be used.

51. Why is process isolation important for system stability?

Answer: Process isolation helps prevent a problem in one application from affecting the entire system. If one process crashes, the operating system can stop the affected process while other processes continue running.

52. Which operating system security mechanism is most directly related to confidentiality? Explain your answer.

Answer: **Access control** is directly related to confidentiality because it determines who is allowed to access specific system resources and information. It helps make sure that sensitive files are only available to authorized users.

53. How do device drivers simplify the work of application developers and users?

Answer: Device drivers act as a connection between the operating system and hardware. They translate operating system requests into instructions that a specific device can understand. This means applications do not have to directly control the hardware electronics.

References

Cavite State University, Department of Information Technology. (n.d.). *Module 3. Operating System Management* [Course module].

Part B — Short Analysis

54. A laptop has 4 GB of RAM and the user opens many applications at once. Explain how the OS may continue operating and why performance could decrease.

Answer: The operating system can use **virtual memory** when the available RAM becomes insufficient. It can move some less-used data between RAM and secondary storage to keep the applications running. However, storage is slower than RAM, so frequent movement of data can cause longer loading times and reduced responsiveness.

55. Two applications attempt to update the same shared data at the same time. What process-management concern is involved, and what type of mechanism can help?

Answer: The concern is **synchronization**. When processes share the same resource, their operations need to be coordinated to prevent conflicts or incorrect results. Mechanisms such as **semaphores** and **mutexes** can help control access to the shared resource.

56. A USB printer is detected, but it does not print correctly. Which device-management component should be investigated first? Explain.

Answer: The **device driver** should be investigated first. The driver translates operating system requests into operations that the printer can understand. If the driver is missing, incompatible, or damaged, the printer may not work correctly.

57. A school computer contains confidential student files. Identify two OS security controls that could protect the files and explain how each helps.

Answer: Two controls are: **Access control** and **Encryption**.

- **Access control** limits which users can access specific files and resources, helping protect confidential information from unauthorized users.
- **Encryption** changes data into an unreadable form unless the correct key is available, helping protect the files if someone gains access to the stored data.

58. A user saves a document and later reopens it. Describe the role of the OS and file system in this process.

Answer: When the document is saved, the operating system works with the **file system** to write the data to storage and keep track of where the file is located. When the user opens it again, the OS uses the file system to find and retrieve the stored data for the application.

References

Cavite State University, Department of Information Technology. (n.d.). *Module 3. Operating System Management* [Course module].

Part C — Performance Task

Scenario: You are part of a small IT support team asked to evaluate a laboratory computer that frequently slows down, has limited RAM, stores many student files, and connects to several USB devices and printers. The school also wants stronger data security.

Prepare a one- to two-page OS Management Improvement Plan that addresses all five areas: process, memory, storage, security, and device management. For each area, identify one observed or possible problem, explain the relevant OS concept, and recommend one practical action.

OS Management Improvement Plan

OS Management Area	Possible Problem	Relevant OS Concept	Recommended Action
Process Management	Too many programs may be running at the same time, which can use a large amount of CPU and memory.	Process management controls running programs and determines how CPU time is shared among processes.	Use Task Manager to check running processes and close unnecessary applications or background processes that are not needed.
Memory Management	The computer has limited RAM, so running many applications at once may reduce available memory and affect responsiveness.	Memory management allocates RAM to programs. The OS can also use virtual memory when physical RAM becomes limited.	Monitor memory usage and close unused applications. If RAM is still insufficient, allow the OS to use virtual memory as needed.
Storage Management	Many student files can consume storage space and make file management more difficult.	Storage management organizes files and folders and uses file systems to manage how data is stored and accessed.	Organize student files into properly labeled folders, remove unnecessary files, and regularly check available storage space.
Security Management	Student files may be exposed to unauthorized access or other security threats.	Security management uses authentication, access control, encryption, and updates to protect systems and data.	Keep the OS updated, use strong authentication, enable the firewall, and use encryption to protect important student files.

Device Management	Several USB devices and printers may cause problems if their drivers are missing or incompatible.	Device management uses device drivers to allow the OS and applications to communicate with hardware.	Check the devices and their drivers through the system's device management tools. Update or reinstall a driver when a device is not working correctly.
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Overall Improvement Plan

The laboratory computer should first be monitored to determine which OS management areas are causing the most problems. **Process and memory management** should be checked because the computer frequently slows down and has limited RAM. Unnecessary applications and processes can be closed, while virtual memory can help when physical memory becomes limited.

For **storage management**, student files should be organized into appropriate folders and unnecessary files should be removed so that storage remains manageable. The IT support team should also monitor available storage space.

For **security management**, the computer should remain updated and use authentication, firewall protection, access control, and encryption where appropriate. These measures can help protect student information from unauthorized access.

Finally, **device management** should be checked because the computer uses several USB devices and printers. The team should verify that the required device drivers are available and working properly.

By addressing these five OS management areas, the laboratory computer can be better organized, protected, and maintained while supporting the needs of students and staff.

References

Cavite State University, Department of Information Technology. (n.d.). *Module 3. Operating System Management* [Course module].

Culminating Assignment — Cloud-Based Operating Systems

Research how a modern cloud-oriented operating system such as Chrome OS manages storage and memory differently from a traditional desktop operating system.

Your submission should include:

- A brief description of the selected cloud-oriented operating system.
- How it handles local storage and cloud storage.
- How memory is managed during normal multitasking.
- At least three differences compared with a traditional desktop OS such as Windows or Linux.
- One advantage and one limitation of the cloud-oriented approach.
- A short conclusion explaining which environment is more appropriate for a school computer laboratory and why.
- A reference list for the sources consulted.

Suggested length: 700–1,000 words or an equivalent 5–7 slide student presentation.

ChromeOS: Storage and Memory Management Compared with Traditional Desktop Operating Systems

Introduction

A cloud-oriented operating system is designed to make strong use of internet-based services and cloud applications. One example is **ChromeOS**, the operating system used by Chromebooks. Google describes ChromeOS as a cloud-first operating system that is designed to provide a secure and manageable computing environment. It can also be used on some existing computers through ChromeOS Flex.

ChromeOS is different from a traditional desktop operating system because many common activities can be connected to online services. However, it still uses the computer's local processor, memory, and storage to run applications and handle files. Therefore, cloud-based computing does not mean that the computer has no local resources.

Local Storage and Cloud Storage

ChromeOS uses both **local storage and cloud storage**. Local storage is used for files saved on the device, such as downloaded files and files that are made available for offline use. ChromeOS provides a Storage Management section where users can see how much space is being used by downloads, offline files, browsing data, applications, and other information.

Cloud storage allows files to be stored remotely and accessed through an internet connection. Google Drive is an example that can be used with ChromeOS. Files can also be made available offline, meaning a copy can be stored on the Chromebook for use without an internet connection. When the device connects to the internet again, changes can synchronize with Google Drive.

This is different from a traditional desktop setup where users may depend more heavily on the computer's internal HDD or SSD for storing applications and files. Windows, for example, requires a storage device with at least 64 GB for Windows 11.

Memory Management During Multitasking

ChromeOS still depends on RAM when several applications, browser tabs, and other tasks are running. When many tabs are open, they can use more memory. Chrome provides a **Memory Saver** feature that can deactivate tabs that are not currently being used to reduce memory use. When an inactive tab is opened again, it can reload.

This relates to the memory-management concepts discussed in the module. The module explains that operating systems manage RAM and may use secondary storage when physical memory becomes limited. It also explains that relying heavily on secondary storage can reduce performance because storage is slower than RAM.

ChromeOS also has mechanisms for handling low-memory situations. Chromium's documentation describes how ChromeOS can reclaim memory from less important background tabs when available memory becomes low.

Differences Between ChromeOS and Traditional Desktop Operating Systems

There are several differences between ChromeOS and traditional desktop operating systems such as Windows or Linux.

First, their approach to storage is different. ChromeOS is designed around cloud services, so users can work with services such as Google Drive while still having local storage for downloads and offline files. Traditional desktop systems such as Windows generally provide a more direct local-storage environment where applications and files can be installed and stored on the computer.

Second, ChromeOS has a strong web-based focus. Many tasks can be performed through web applications and Google services. Windows and Linux can also use web applications, but they generally provide broader support for locally installed desktop software.

Third, ChromeOS has built-in security features designed around its cloud-focused environment. ChromeOS uses features such as Verified Boot, a read-only operating system, sandboxing, automatic updates, and data encryption.

Fourth, device management can be centralized. ChromeOS devices can be managed through the Google Admin console, allowing administrators to apply policies and manage multiple devices.

Advantage and Limitation

One advantage of ChromeOS is **easy access to cloud-based files and services.** Students can work with files stored in Google Drive and can access them from different devices when connected to the internet. ChromeOS also provides automatic security and update features that can reduce some maintenance work.

One limitation is that **some tasks depend on internet access and cloud-compatible applications.** Although ChromeOS supports offline work for certain files and applications, Google notes that some apps and services may not work when the device is offline. Users who need specialized desktop applications may also need alternative solutions or web versions of those applications.

Conclusion

For a school computer laboratory, ChromeOS can be an appropriate environment when the laboratory mainly uses web applications, cloud storage, and online learning tools. Its cloud-focused design can make file access and device management convenient, while built-in security features can help protect the computers and their data.

However, a traditional desktop operating system such as Windows may be more suitable when students need specialized locally installed software that is not available through web applications. Therefore, the more appropriate environment depends on the school's software requirements, internet availability, storage needs, and the type of activities students perform.

Overall, ChromeOS shows how an operating system can still perform the same basic management functions discussed in the module, such as memory, storage, security, and device management, while using a more cloud-oriented approach.

References

Cavite State University, Department of Information Technology. (n.d.). *Module 3. Operating System Management* [Course module].

Google. (n.d.). ChromeOS. <https://chromeos.google/>

Google. (n.d.). *Free up space on your Chromebook*. Google Chromebook Help. <https://support.google.com/chromebook/answer/1061547>

Google. (n.d.). *Personalize Chrome performance*. Google Chrome Help. <https://support.google.com/chrome/answer/12929150>

Google. (n.d.). *Use Google Drive files offline on your Chromebook*. Google Chromebook Help. <https://support.google.com/chromebook/answer/2809731>

Microsoft. (n.d.). *Windows 11 system requirements*. Microsoft Support. <https://support.microsoft.com/en-us/windows/windows-11-system-requirements>

Student Reflection

59. Which OS management function was easiest for you to understand? Why?

Answer: The easiest function for me to understand was **storage management**. I chose this because, as a student, documents are already an essential part of my devices. Storage management helped me to understand how these documents can be organized, such as changing the file names, fixing the order of files, storing and removing files, and making my files available whenever I need them.

60. Which function do you think has the greatest effect on everyday system performance? Defend your answer.

Answer: For me, **memory management** has one of the biggest effects on everyday performance because the amount of available RAM affects how many applications can run smoothly at the same time. When there's insufficient memory, the computer may depend more on other storage options to keep programs running, which can lead to slower system performance and cause some applications to become unresponsive or crash. Furthermore, this situation affects the speed of communication between the hardware and the applications.

61. Which security practice from the module can you apply immediately to your own device?

Answer: The security practice I can apply immediately is **using proper authentication**. I can make sure that my device is protected with secure sign-in methods so that other people cannot easily access my files and personal information.

62. Give one real-world example in which process, memory, storage, security, and device management work together during a single user task.

Answer: One real-world example is **attending an online class**. The meeting application runs as a process and uses memory for video, audio, and other activities. Storage management handles files or recordings saved on the device. Security helps protect the account and information from unauthorized access. Device management allows the camera, microphone, speakers, and other hardware to communicate with the application. These functions work together during the online class.